

Trupti V Hubli

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LinkedIn | GitHub

Professional Summary

AI and Data Science graduate with hands-on experience in data science, software development, backend development, machine learning, Generative AI, and full-stack application development. Skilled in Python, SQL, JavaScript, FastAPI, React.js, REST APIs, data analysis, machine learning, NLP, and RAG. Experienced in building data-driven solutions, AI-powered applications, real-time monitoring systems, and scalable software projects through internships, academic projects, and research experience.

Technical Skills

Programming: Python, SQL, JavaScript, R

Data Analysis & Visualization: Pandas, NumPy, Scikit-learn, EDA, Data Cleaning, Power BI, Tableau, Matplotlib, Seaborn

AI & Machine Learning: Machine Learning, Deep Learning, Generative AI, LLMs, NLP, RAG, Prompt Engineering, Anomaly Detection, Predictive Analytics, Statistical Modeling, Feature Engineering, Computer Vision

Software Development: FastAPI, React.js, REST APIs, Frontend Development, Backend Development, Responsive Web Design, Git, GitHub, Version Control

Frameworks & Libraries: Hugging Face Transformers, TensorFlow, Keras, PyTorch, OpenCV, gTTS

Databases & Tools: MySQL, PostgreSQL, MongoDB, ChromaDB, Jupyter Notebook, Google Colab, VS Code

Experience

VisionAstraa EV Academy

Data Science Intern

Feb 2026 – May 2026

Bengaluru, India

- Collected, cleaned, and processed EV charging network datasets from multiple sources to support machine learning applications.
- Developed and evaluated anomaly detection models using Isolation Forest and XGBoost to classify spoofing and intrusion attempts.
- Performed data preparation, segmentation, and feature engineering workflows to support predictive analytics and model performance optimization.
- Built real-time monitoring dashboards and RESTful APIs using FastAPI, documenting code and model workflows for reproducibility and maintainability.
- Communicated model findings and operational insights to cross-functional stakeholders through visualizations and structured reporting.

Yuva Intern

Virtual R Data Analyst

Jan 2026 – Feb 2026

Bengaluru, India

- Built multiple linear regression models to predict fuel efficiency using R and ggplot2, achieving $R^2 = 0.83$ and Pearson $r = 0.868$ ($p < 0.001$).
- Conducted correlation analysis and statistical modeling to identify key predictors and support data-driven decision making.
- Created bar charts, histograms, scatter plots, and boxplots to visualize insights and communicate findings.
- Applied data cleaning, statistical modeling, and visualization techniques to derive actionable insights from structured datasets.

Deloitte – Forage

Data Analytics Job Simulation

Dec 2025

Virtual Experience

- Conducted exploratory data analysis on business datasets to identify trends, anomalies, and actionable insights.
- Cleaned, transformed, and analyzed structured datasets using data analytics methodologies.
- Created visualizations and dashboards to communicate findings and support data-driven business decisions.

Projects

DocMind – LLM-Powered RAG Document Assistant

Python, FastAPI, ChromaDB, Hugging Face

- Developed an LLM-powered Retrieval-Augmented Generation (RAG) assistant for document ingestion, semantic retrieval, and question answering.
- Designed RESTful APIs using FastAPI and integrated vector embeddings and ChromaDB for intelligent document search.
- Integrated frontend, backend, and database components to deliver an end-to-end AI application.

OCPD Spoofing Detection

Python, FastAPI, XGBoost

- Developed a machine learning solution to detect spoofing and intrusion attempts in EV charging communication networks.
- Performed data preprocessing, feature engineering, anomaly detection, and model evaluation for attack classification.
- Built FastAPI-based backend services for real-time monitoring and alert generation.

AI-Based Driver Monitoring and Collision Avoidance System

Python, OpenCV, TensorFlow

- Developed a real-time computer vision system for driver fatigue detection and collision avoidance using OpenCV and deep learning techniques.
- Implemented facial and behavioral monitoring algorithms to identify drowsiness, distraction, and unsafe driving patterns.
- Built a live analytics dashboard using Python and Matplotlib and integrated real-time video processing and alerts.

Smart Sign – CNN-Based Traffic Sign Recognition

Python, TensorFlow, Keras, OpenCV, gTTS

- Developed an end-to-end CNN-based computer vision solution to classify traffic signs from road images.
- Implemented image preprocessing, data augmentation, model training, and hyperparameter tuning using TensorFlow and Keras.
- Integrated Google Text-to-Speech (gTTS) to generate real-time audio responses for detected traffic signs.
- Evaluated model performance using accuracy metrics, confusion matrices, and classification reports; published research findings in a peer-reviewed journal.

ClosetCurated – Fashion Discovery Web Application

HTML, CSS, JavaScript, GitHub Pages

- Developed a responsive web application for curated fashion and styling recommendations.
- Designed interactive user interfaces using HTML, CSS, and JavaScript with responsive web design principles.
- Managed deployment using GitHub Pages and version control through Git and GitHub.

Education

BGS College of Engineering and Technology

Dec 2022 – May 2026

Bachelor of Engineering in Artificial Intelligence and Data Science

Bengaluru, India

CGPA: 6.9/10

Certifications

- Artificial Intelligence Fundamentals – Great Learning
- Introduction to Natural Language Processing – Great Learning
- Implementing Deep Learning – Infosys Springboard
- Techniques for Big Data Analytics – Infosys Springboard
- Statistical Analysis and Modeling in R – Infosys Springboard
- Data Visualization with Tableau Foundation
- Data Visualization with Tableau Pro
- Power BI for Beginners